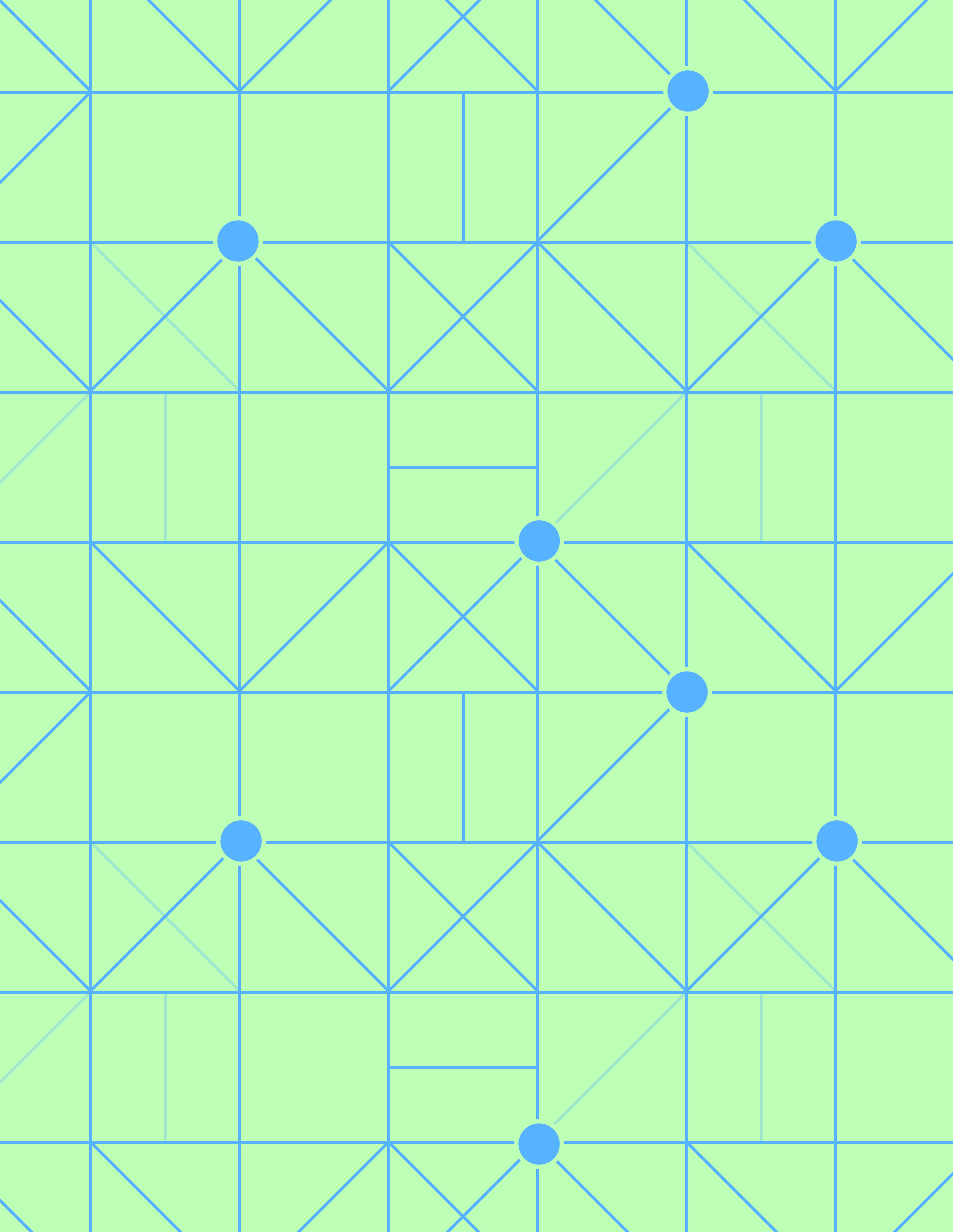

SIMPLEX METHOD GOES WRONG ...

- TIE FOR EBV
(i.e., TWO VARIABLES HAVE THE SAME COEFFICIENT IN THE OBJECTIVE FUNCTION AND THEY WOULD PRODUCE THE SAME - HIGHEST - INCREASE)

=> PICK ARBITRARILY

- TIE FOR LBV
(i.e., TWO VARIABLES GIVE THE SAME MINIMUM RATIO TEST MEANING THAT THE CORRESPONDING CPF SOL ARE "AT THE SAME DISTANCE" FROM THE CURRENT CPF)

=> WHICH ONE YOU PICK IS RELEVANT FOR THE FOLLOWING STEPS
MOREOVER BOTH WILL GO TO ZERO WHEN THE EBV ENTERS (EVEN THE ONE THAT DOESN'T LEAVE THE BASIS)

[BASIC VARIABLES WITH ZERO VALUE ARE CALLED DEGENERATE

AS A CONSEQUENCE, IN THE NEXT STEP THE VALUE OF THE OBJECTIVE FUNCTION WILL NOT INCREASE

- STUCK [SOME STEPS ARE NEEDED TO GO FROM THE CURRENT CPF TO THE NEXT
- CYCLE [ALWAYS THE SAME CPF ARE VISITED AND THEY REPEAT THEMSELVES

- UNBOUNDED OBJECTIVE FUNCTION
(i.e. NO LBV)

[NO EBV CAN NOT HAPPEN BECAUSE OF OPTIMALITY

⇒ THIS HAPPEN WHEN THE EBV CAN BE INCREASED INDEFINITELY WITHOUT GIVING A NEGATIVE VALUE TO THE BV (i.e., WITHOUT VIOLATING THE CONSTRAINTS)

- MULTIPLE OPTIMAL SOL
(i.e. CONVEX COMBINATION OF OPTIMAL CPF SOL)

⇒ SIMPLEX METHOD ALLOWS TO FIND ALL OF THE OPTIMAL CPF SOL

⇒ AT LEAST ONE OF THE N.BV HAS COEFFICENT EQUAL ZERO IN THE OBJECTIVE FUNCTION EQ SO INCREASING THIS VARIABLE WILL NOT CHANGE THE VALUE OF THE OBJECTIVE FUNCTION

THE OTHER CPF CAN BE FOUND BY CHOOSING SUCH NBV AS EBV